

# Red Team vs Pentest vs Adversary Emulation

## The spectrum

|             | Pentest                  | Red Team                                 | Adversary Emulation                 |
|-------------|--------------------------|------------------------------------------|-------------------------------------|
| Goal        | Find vulnerabilities     | Test DETECTION + RESPONSE                | Replicate a specific APT's behavior |
| Timing      | Typically days           | Weeks-months, stealthy                   | Focused on a named actor TTP set    |
| Stealth     | Minimal (scans are fine) | High (evade the defenders)               | High                                |
| Deliverable | Vuln report              | 'How well did you detect?' report + gaps | ATT&CK-mapped chain                 |

## What red teaming tests

Detection: did the SOC/EDR/SIEM catch the operation?

Response: could they contain, investigate, and recover?

Human: phishing that worked, MFA gaps, policy holes

OPSEC: can the attacker sustain independence (C2 survives)?

## Adversary emulation in practice

Choose a real threat actor (from a CTI report: e.g., a specific APT)

Map their known TTPs to ATT&CK techniques

Execute a chain that replicates their sequence

Compare results vs the mapping – where did defense hold?

## Purple team (people to befriend)

Red + Blue run TOGETHER: red forces a technique, blue shares detection learnings immediately

Purpose: rapid improvement, not proof

The modern default, honestly better than 'us vs them'

## Scoping realism

RoE must define: can you hit backups? cloud production? can you trigger pager? (usually yes, carefully)

Goals in writing: "detect the operation within X hours" / "exfil simulated data in a sandbox"

Kill switches: emergency stop-word + 'stop if production harmed'

Red teaming is a detection-and-response bootcamp with a budget. Depth, stealth, and evaluation – not vuln counting – are the metrics.

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